



HuBMAP in the CFDE: Discovering, Visualizing, and Reusing Multimodal Omics Data

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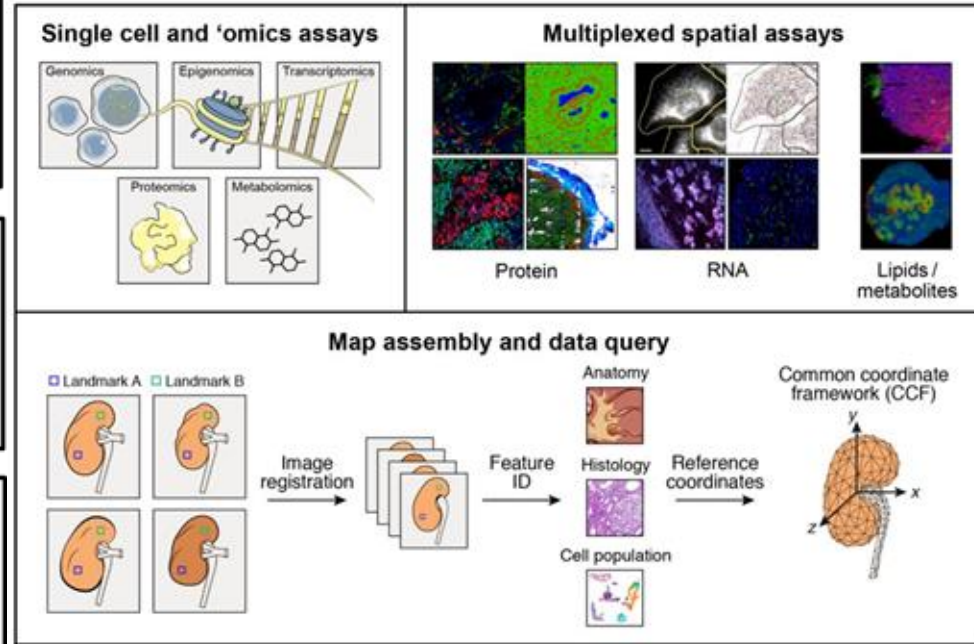
Pittsburgh Supercomputing Center

Human BioMolecular Atlas Program (HuBMAP)

"While we broadly know how cells are organized in most tissues, a comprehensive understanding of the cellular and molecular states and interactive networks resident in the tissues and organs [...] is lacking."

"The specific three-dimensional organization of different cell types, together with the effects of cell–cell and cell–matrix interactions in their natural milieu, have a profound impact on normal function, natural ageing, tissue remodelling, and disease progression in different tissues and organs."

"HuBMAP is an important part of the international mission [and] aspires to help to build a foundation by generating a high-resolution atlas of key organs [...] as well as acting as a key resource for new contributions in the growing fields of tissue biology and cellular ecosystems."

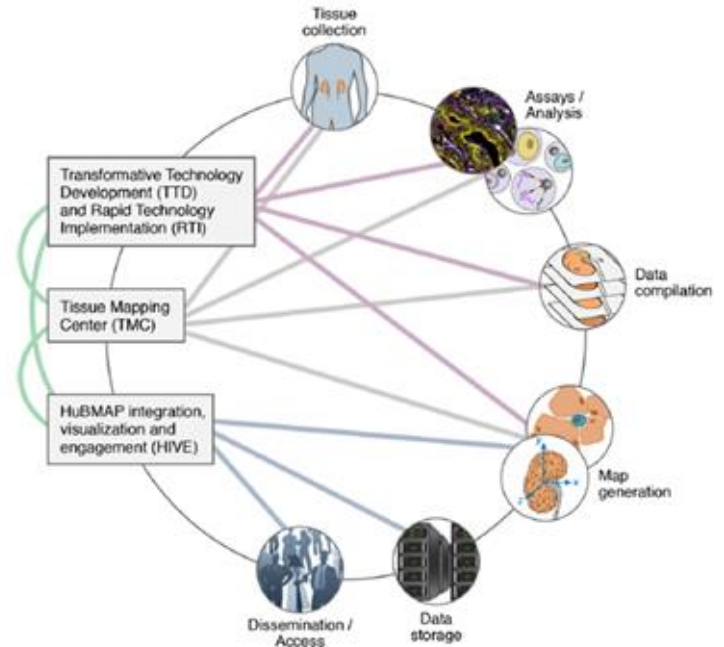


HuBMAP Vision and Goals

Goal: Catalyze the development of a framework for mapping the human body at single cell resolution

By:

1. Accelerating development of next generation tools and techniques
2. Generating foundational 3D human tissue maps
3. Establishing an open data platform
4. Collaborating with the research community
5. Supporting pilot projects that demonstrate value of HuBMAP resources





The HuBMAP Consortium

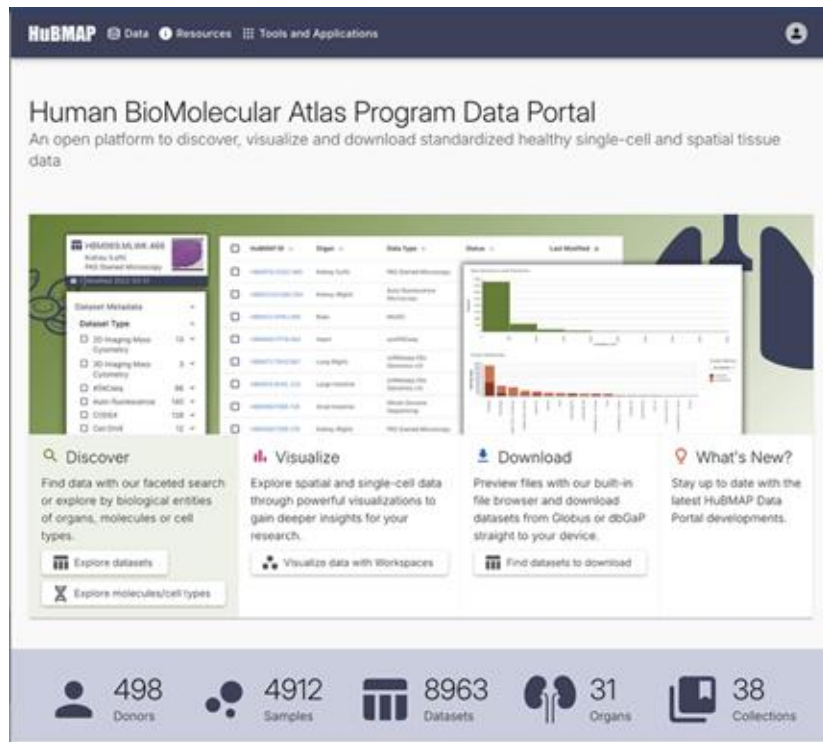
Key Features

- Free, **open access** to non-identifiable data, tools and technologies developed
- **Healthy** tissue
 - Diversity in samples: sex, ethnicity, age
- Expansive, accessible **three-dimensional** molecular and cellular atlas of the human body
- Culture of **openness and sharing** using **team science-based** approaches
 - Protocols.io, github
- **Diverse expertise in funded members**
 - molecular, cellular, developmental, and computational biologists, measurement experts, clinicians, pathologists, anatomists, biomedical and software engineers, and computer and data information scientists
- Coupled with no-cost access leading **HPC resources** and cloud burstability.

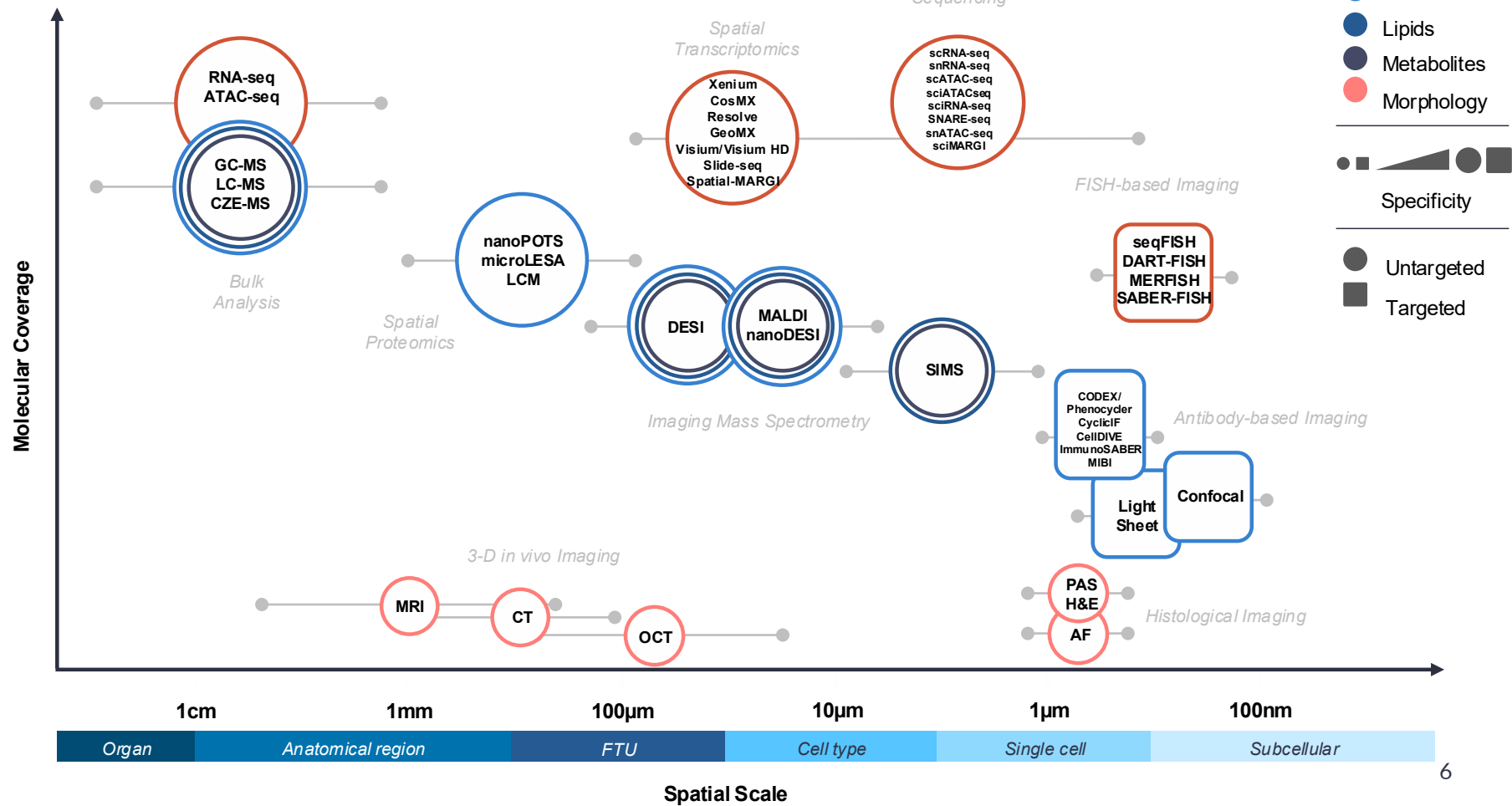
The HuBMAP Data Portal:

<https://portal.hubmapconsortium.org/>

Serves as a **repository** for multi-modal spatial and single-cell data from **healthy human tissues** to generate a high-resolution atlas of the human body



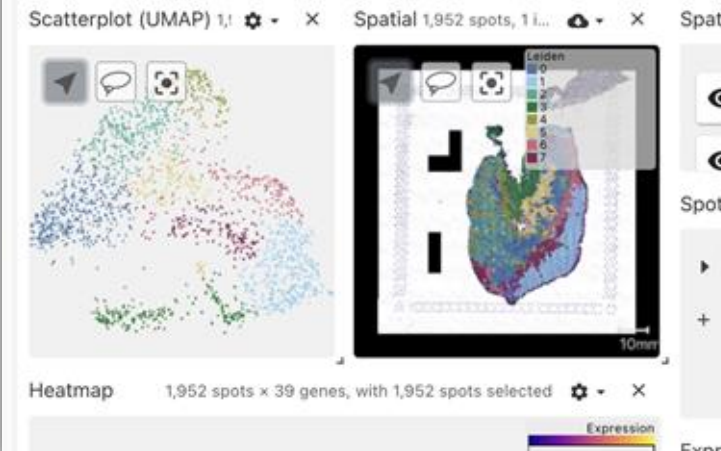
HuBMAP Technologies



HuBMAP Analysis Pipelines

- HuBMAP analysis pipelines support multiple assay families:
 - Single-cell / single-nucleus sequencing
 - sc/snRNA-seq
 - sc/snATAC-seq
 - multiome RNA + ATAC
 - Spatial transcriptomics
 - Visium
 - Slide-seq
 - Imaging and spatial proteomics
 - PhenoCycler
 - CODEX
 - CellIDIVE
 - MIBI
 - FISH-based assays
- These pipelines help produce standardized, reusable outputs for Portal discovery, visualization, and downstream analysis.

Visualization



HBM476.VHDB.268

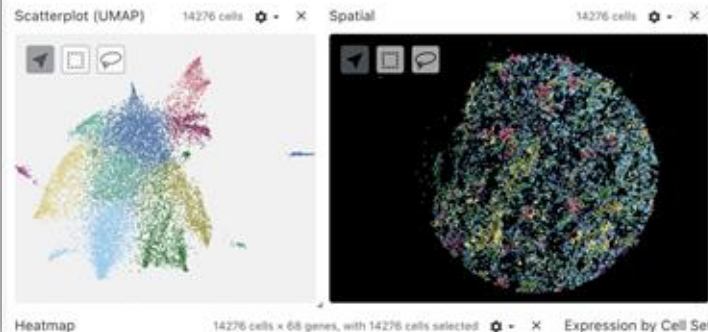
Slide-seq [Salmon] | Kidney (Right)

Published | Public Access

Publication Date
2022-02-14

Modification Date
2022-02-14

Visualization

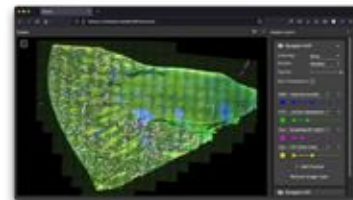
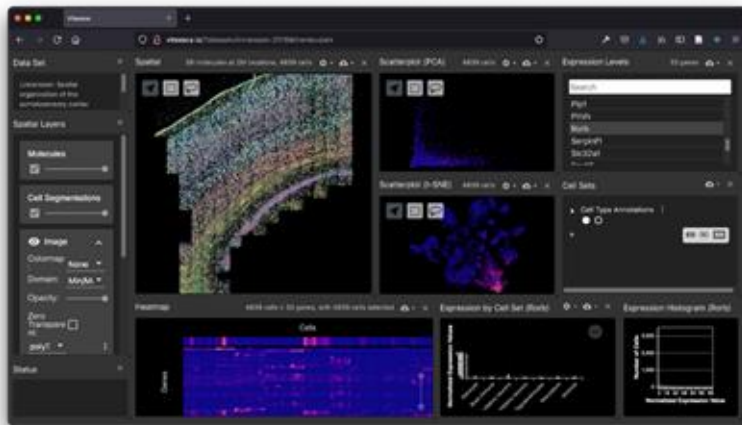


Vitessce

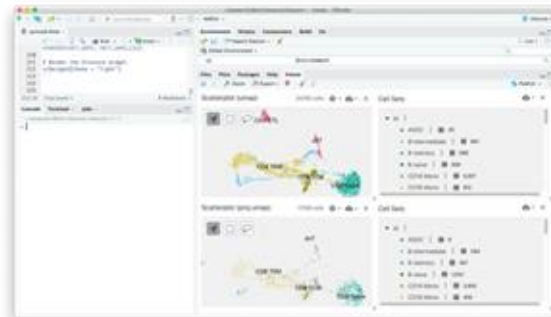
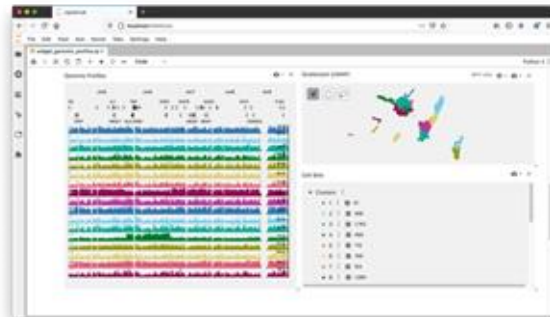
Visual integration tool for exploration of spatial single cell experiments

<https://vitessce.io/>

- Modular interactive visualization framework for single-cell data
- Lack of specialized server infrastructure
- Coordination of common entities across views
- Support for data analysis environments (Python/R)
- Support for common single-cell data structures/file formats



Web browser





HuBMAP Omics Data: Discovery and Reuse

Key Portal Requirements

Meet needs of a breadth of users

Portal user interfaces must serve a wide range of user audiences, needs, and workflow entry points particularly for features related to:

- **Search** and **Discovery** of data
- **Visualization** and **Analysis** of data directly in the portal
- **Download** of large collections of data

Portal tools and infrastructure must:

- **Support data** from multiple individuals at a range of anatomical scales (organ to subcellular)
- Follow findable, accessible, interoperable, and reusable **(FAIR) data principles**
- Maintain accuracy and **reproducibility**; protect data with appropriate levels of **privacy**
- Follow **community-developed data standards** for algorithms to run at scale
- Ensure **long-term availability** of open tools, data, and infrastructure

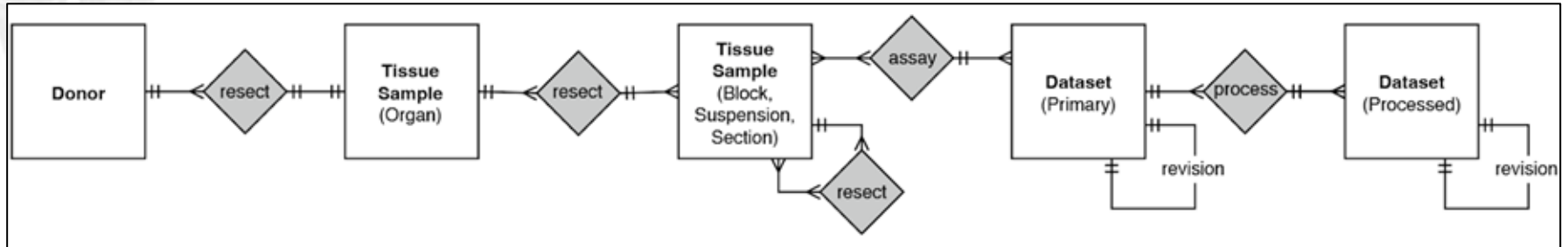
Data on the HuBMAP Data Portal

Biological entities

Donor > organ > sample > datasets (raw and processed)

Provenance is represented on the portal

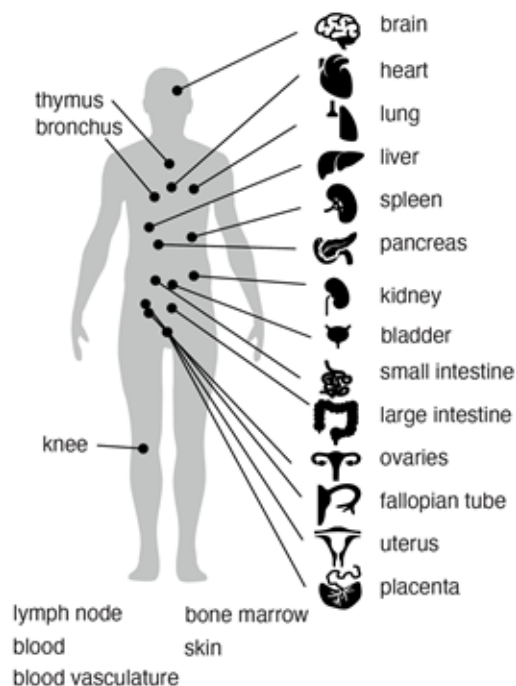
- Tracks entire lifecycle of biological samples and data
- Where data originated, how they've been processed, all associated metadata



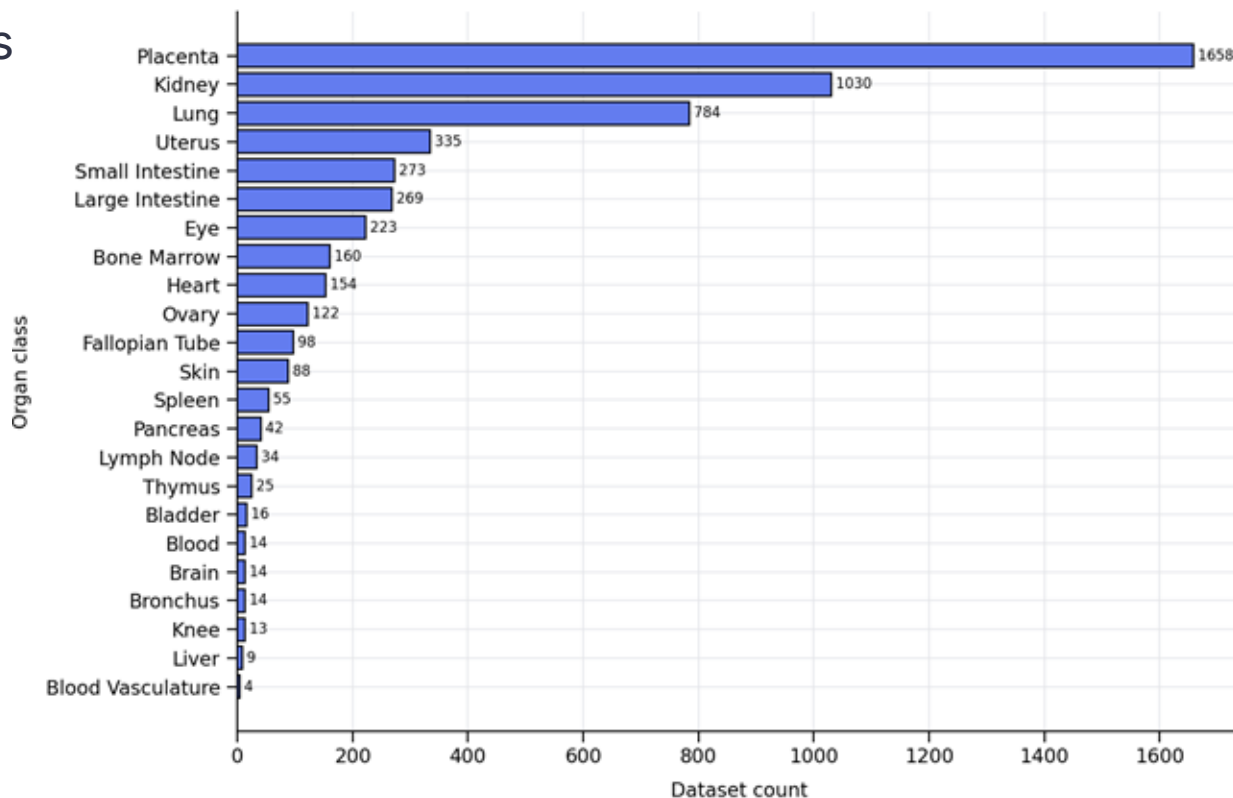
Organs

28 organ classes

400+ unique donor organs

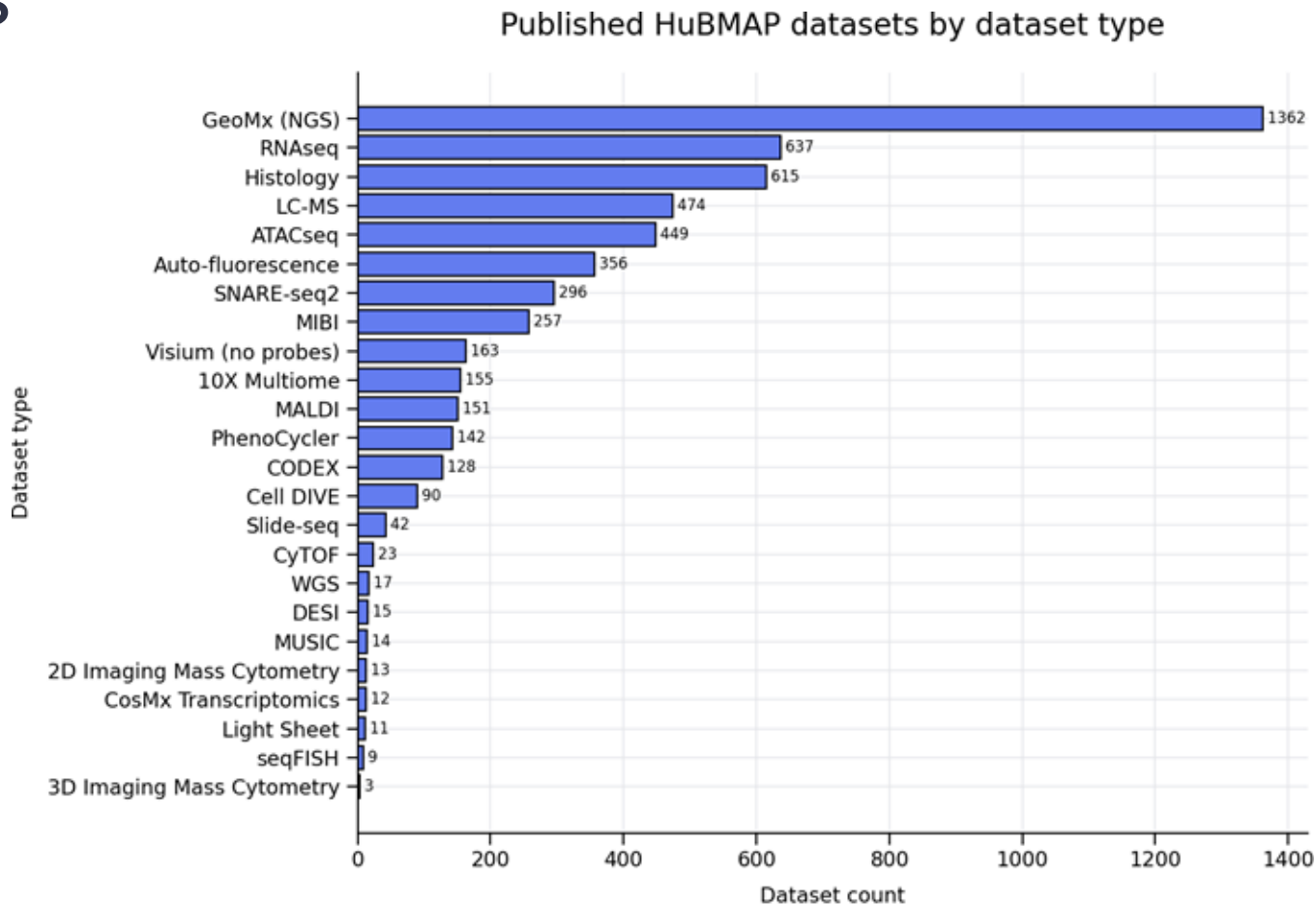


Published HuBMAP datasets by organ class

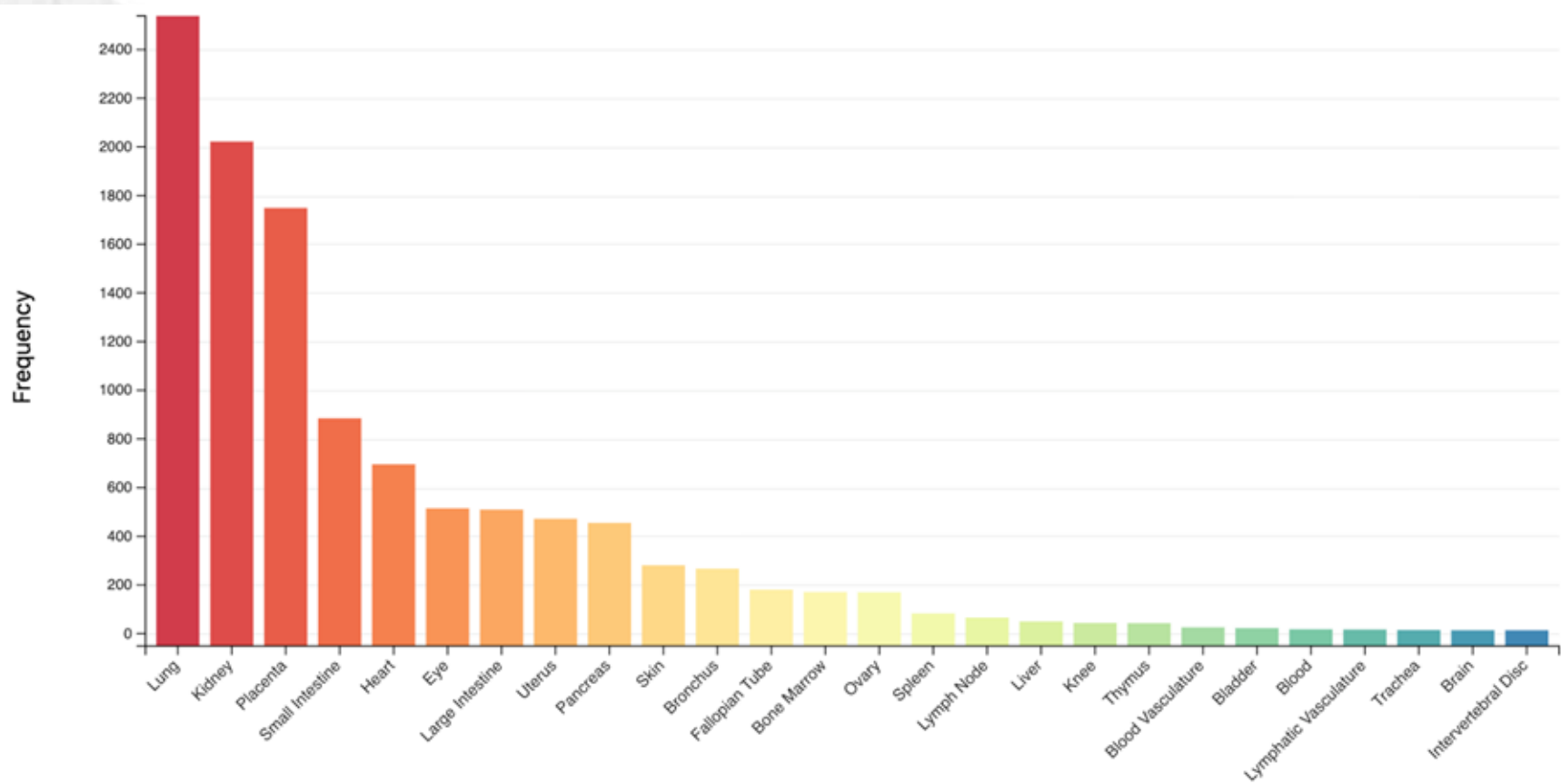


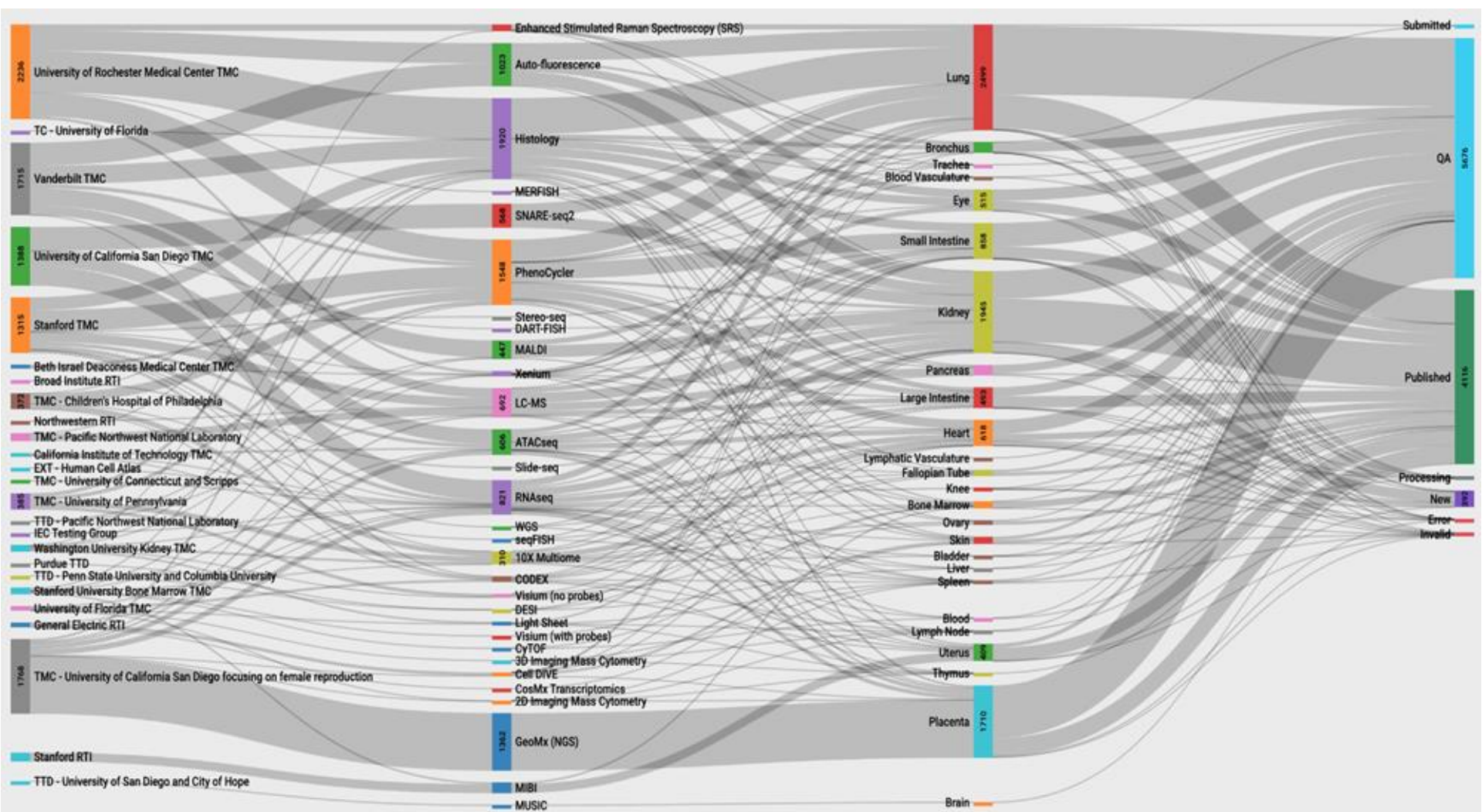
Data Types

24 currently
on the portal

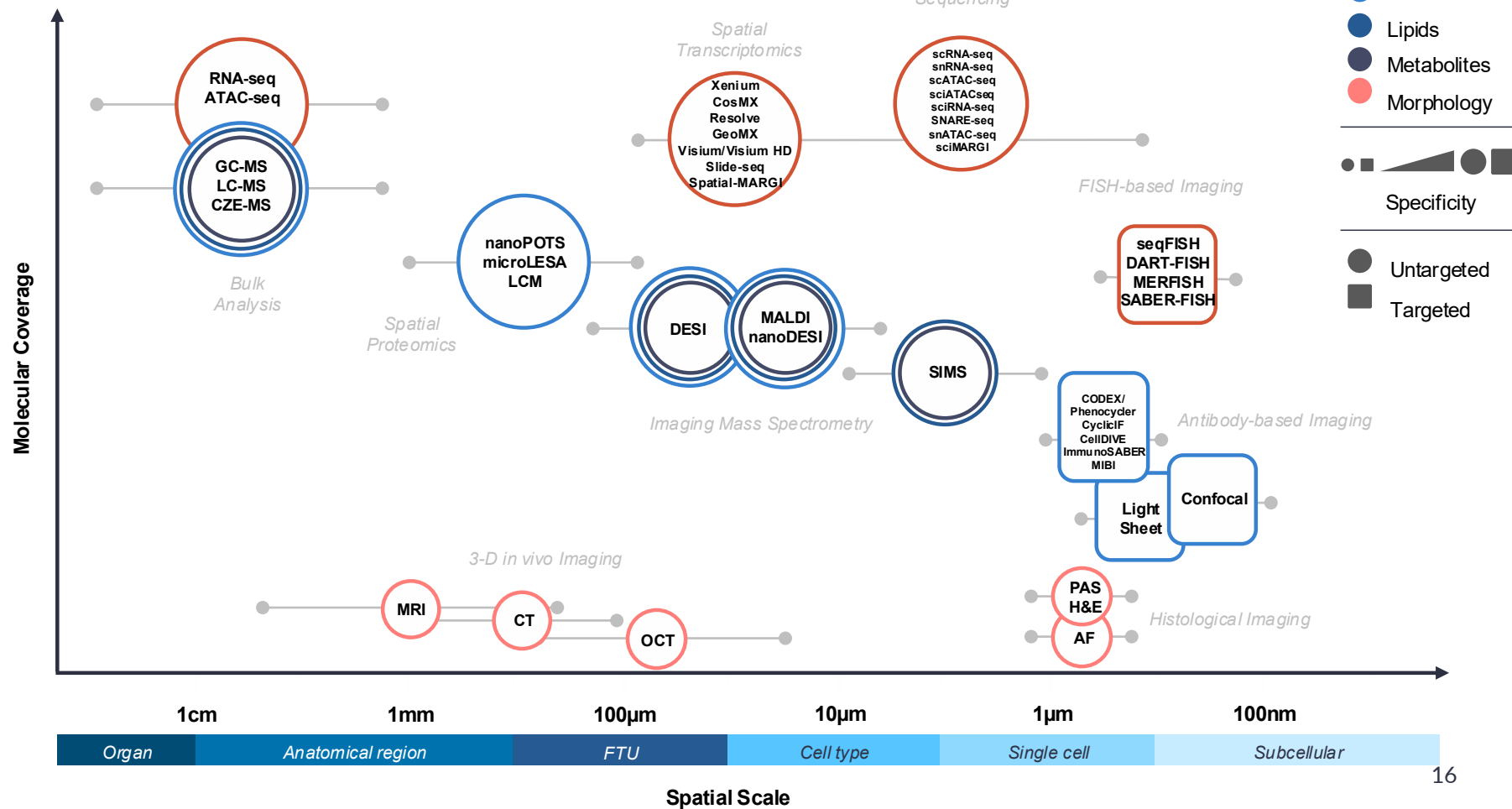


Current Snapshot of Published/Soon-to-be Published HuBMAP Data





HuBMAP Technologies

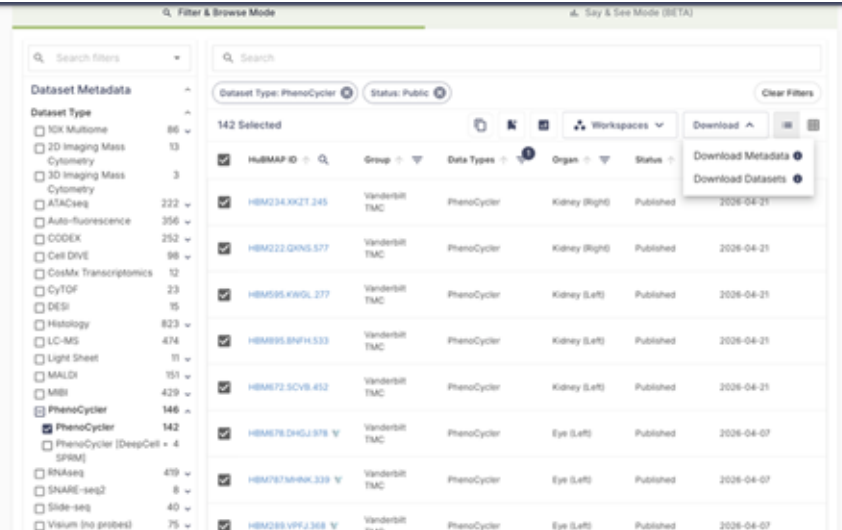


Examples of HuBMAP Datasets (Live Demos)

- Faceted search
- Example 1: Visium (no probes) - spatial transcriptomics (where RNA expression occurs in tissue)
 - Dataset [HBM588.GLJR.289](#)
- Example 2: PhenoCycler - spatial proteomics (where protein markers occur in tissue)
 - Dataset [HBM849.GZFL.769](#)
- Items of note:
 - Metadata
 - Provenance
 - Visualizations
 - Files/data products/data access - dbGaP access for sequencing data
 - Protocols and workflow details

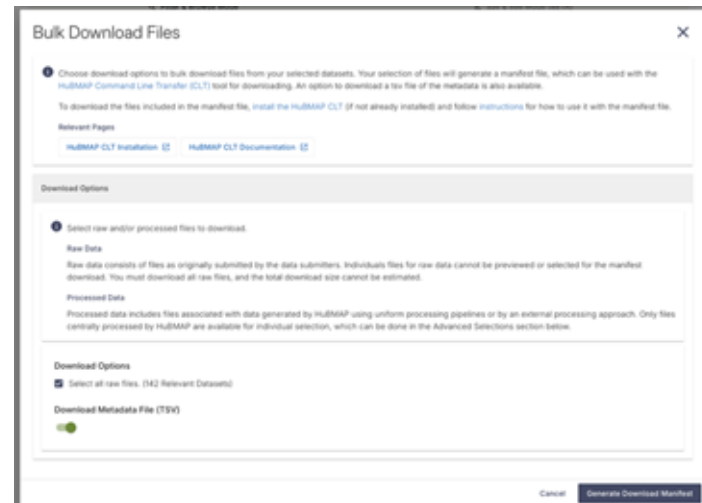
Reusing HuBMAP Data

- Downloading HuBMAP data
 - Per the previous examples, you can use Globus to download non-sequencing data and [dbGaP](#) for raw datasets containing human sequencing data
- Bulk downloading HuBMAP data
 - HuBMAP offers a Command Line Transfer (CLT) utility for downloading multiple datasets at a time
 - This process uses a manifest.txt file generated from a faceted dataset search on the portal
 - [Installing the CLT](#)
 - [Using the CLT](#)



The screenshot shows the HuBMAP portal interface. On the left is a sidebar with 'Dataset Metadata' and a list of dataset types with counts. The main area displays a table of 142 selected datasets. The table has columns for 'Dataset ID', 'Group', 'Data Types', 'Organ', and 'Status'. A 'Download' button is visible in the top right corner of the table area.

Dataset ID	Group	Data Types	Organ	Status
HBM234.KKZT.245	Vanderbilt TMC	PhenoCycler	Kidney (Right)	Published
HBM222.QNNS.577	Vanderbilt TMC	PhenoCycler	Kidney (Right)	Published
HBM595.KWGL.277	Vanderbilt TMC	PhenoCycler	Kidney (Left)	Published
HBM895.BNPH.533	Vanderbilt TMC	PhenoCycler	Kidney (Left)	Published
HBM672.SCVB.452	Vanderbilt TMC	PhenoCycler	Kidney (Left)	Published
HBM678.DHGL.375	Vanderbilt TMC	PhenoCycler	Eye (Left)	Published
HBM787.MHKL.339	Vanderbilt TMC	PhenoCycler	Eye (Left)	Published
HBM288.VPFL.368	Vanderbilt TMC	PhenoCycler	Eye (Left)	Published



The 'Bulk Download Files' dialog box provides instructions for downloading multiple datasets. It includes a section for 'Download Options' where users can select raw or processed files. The 'Download Metadata File (TSV)' option is currently selected. At the bottom, there are 'Cancel' and 'Generate Download Manifest' buttons.

Bulk Download Files

Choose download options to bulk download files from your selected datasets. Your selection of files will generate a manifest file, which can be used with the [HuBMAP Command Line Transfer \(CLT\)](#) tool for downloading. An option to download a file of the metadata is also available.

To download the files included in the manifest file, [install the HuBMAP CLT](#) (if not already installed) and follow [instructions](#) for how to use it with the manifest file.

Relevant Pages

[HuBMAP CLT Installation](#) [HuBMAP CLT Documentation](#)

Download Options

Select raw and/or processed files to download.

Raw Data

Raw data consists of files as originally submitted by the data submitters. Individual files for raw data cannot be previewed or selected for the manifest download. You must download all raw files, and the total download size cannot be estimated.

Processed Data

Processed data includes files associated with data generated by HuBMAP using uniform processing pipelines or by an external processing approach. Only files centrally processed by HuBMAP are available for individual selection, which can be done in the Advanced Selections section below.

Download Options

Select all raw files. (142 Relevant Datasets)

Download Metadata File (TSV)

☒ ☐

Cancel Generate Download Manifest

Come Visit Us!

HuBMAP Core Resources

- [HuBMAP Data Portal](#)
- [HuBMAP Dataset Search](#)
- [HuBMAP Publications](#)
- [HuBMAP Documentation](#)

Visualization and Analysis

- [Vitessce](#)
- [HuBMAP Workspaces](#)
- [Azimuth for HuBMAP](#)
- [Human Reference Atlas](#)

The screenshot shows the HuBMAP Data Portal interface. At the top, the header includes the HuBMAP logo, navigation links for Data and Resources, and a user profile icon. Below the header, the main title is "Human BioMolecular Atlas Program Data Portal" with the subtitle "An open platform to discover, visualize and download standardized healthy single-cell and spatial tissue data". The main content area features a large image showing a 3D visualization of a tissue section with various colored cells and structures. To the left of this image is a sidebar with a search and filter panel. Below the main image, there are four columns of content: "Discover" (Find data with our faceted search or explore by biological entities of organs, molecules or cell types), "Visualize" (Explore spatial and single-cell data through powerful visualizations to gain deeper insights for your research), "Download" (Preview files with our built-in file browser and download datasets from Globus or doGAP straight to your device), and "What's New?" (Stay up to date with the latest HuBMAP Data Portal developments). At the bottom, there is a footer with statistics: 310 Donors, 3118 Samples, 4742 Datasets, 31 Organs, and 20 Collections.

HuBMAP Data Resources

Human BioMolecular Atlas Program Data Portal

An open platform to discover, visualize and download standardized healthy single-cell and spatial tissue data

Discover
Find data with our faceted search or explore by biological entities of organs, molecules or cell types.
Explore datasets
Explore molecules/cell types

Visualize
Explore spatial and single-cell data through powerful visualizations to gain deeper insights for your research.
Visualize data with Workspaces

Download
Preview files with our built-in file browser and download datasets from Globus or doGAP straight to your device.
Find datasets to download

What's New?
Stay up to date with the latest HuBMAP Data Portal developments.

310 Donors 3118 Samples 4742 Datasets 31 Organs 20 Collections

For any questions, email help@hubmapconsortium.org